LUNDS UNIVERSITET

Tentamensskrivning
Topologi
Lördag den 24 april 2021
Skrivtid: 08.00 – 13:00

Matematikcentrum
Matematik NF

Använd LaTeX, eller skriv tydligt för hand med svart penna på vanligt vitt papper. Lös inte flera uppgifter på samma sida. Skriv ditt namn och personnummer på varje sida.

Lämna in antingen en LaTeX-pdf-fil, eller skanna in/fotografera dina handskrivna sidor och samla dessa i en enda pdf-fil. Kontrollera att din fil är läslig, och lämna in via CANVAS. Om du saknar möjlighet att lämna in via CANVAS så kan du skicka dina lösningar till Sandra Pott sandra.pott@math.lu.se via email. Endast inlämningar gjorda innan 14:00 lördag 24 april kommer att accepteras.

Inget samarbete är tillåtet för att lösa tentamensproblemen. I dina lösningar tillåts du använda/hänvisa till resultat som formuleras/bevisas i föreläsningsanteckningarna och övingarna. Ange tydliga hänvisningar till respektive resultat (t.ex. "Theorem on the characterisation of compact sets in metric spaces" eller "Lemma 2 in Lecture 13"). Om du använder andra resultat (till exempel från kursboken) så måste du inkludera ett fullständigt bevis. Ge alltid klara och noggranna motiveringar, och rita en figur när detta är lämpligt.

Notera även att du får använda påståenden i tidigare delar av en uppgift för att lösa de senare delarna, även om de inte löst de tidigare.

Please use either LaTeX or write with clear handwriting and black pen on plain white paper. Please treat at most one exercise per page, and put your name and personal number on each page.

Please submit either a LaTeX-pdf file or scan/photograph your handwritten pages, combining your submission in one single pdf file. Check that your submission file is clearly readable, and submit in via CANVAS. In case that you have not the possibility of submitting via CANVAS, you can send it to Sandra Pott sandra.pott@math.lu.se via email. Only submissions made until 14:00 on Saturday April 24 will be accepted.

No cooperation is allowed to solve the exam problems. In your solutions, you may use/cite results that are stated and/or proved in the lecture notes or the exercises for the seminars. Please include a clear reference to the respective result (e.g "Theorem on the characterisation of compact sets in metric spaces" or "Lemma 2 in Lecture 13"). If you use other result (for example from the textbooks), you have to include a full proof. Give always clear careful motivations, including sketches where appropriate.

Note also that you can use the statements of the earlier parts of a problem to solve the later parts, even if you have not solved the earlier part.

1. 1. Let $\mathbb{R}^4$ be equipped with the standard topology induced by the Euclidean metric and let $f : \mathbb{R}^4 \to \mathbb{R}$ be the function given by

$$f(x_1, x_2, x_3, x_4) = x_1^2 - x_2^2 + x_3^2 - x_4^2.$$

Put $A = f^{-1}(\{1\})$, $B = A \cap \{x \in \mathbb{R}^4 : x_2^2 + x_4^2 = 1\}$ and $C = f^{-1}((0, \infty)) \cap \{x \in \mathbb{R}^4 : x_2 = x_3 = x_4 = 0\}$. Prove whether the following statements are true or false.

Var god vänd!

a) A is compact,

b) B is compact,

c) B is path-connected,

d) C is connected.

2. Let

$$\mathcal{B} = \{[a, b) : a, b \in \mathbb{R}, a < b\}.$$

a) Show that $\mathcal{B}$ is the basis of a topology on $\mathbb{R}$.

b) Prove that the topology $\mathcal{T}_{\mathcal{B}}$ generated by $\mathcal{B}$ is finer than the standard topology $\mathcal{T}$ on $\mathbb{R}$.

c) Prove that $\mathbb{R}$ is totally disconnected in the topology $\mathcal{T}_{\mathcal{B}}$ generated by $\mathcal{B}$.

3. Let $(X, \mathcal{T})$, $(Y, \mathcal{T}')$ be topological spaces, and let $f, g : X \to Y$ be two continuous maps. Suppose that Y is Hausdorff.
Show that $\{x \in X : f(x) = g(x)\}$ is closed in $(X, \mathcal{T})$.

4. Let $\mathcal{T}_f$ be the co-finite topology on $\mathbb{R}$, given by

$$U \in \mathcal{T}_f, \text{ if and only if } \mathbb{R} \backslash U \text{ finite or } U = \emptyset.$$

For each $N \in \mathbb{N}$, let $A_N \subset \mathbb{R}$ be given by $A_N = \{\frac{1}{n} : n \in \mathbb{N}, n \leq N\} \cup \{0\}$. Moreover, let $A \subset \mathbb{R}$ be given by $A = \{\frac{1}{n} : n \in \mathbb{N}\} \cup \{0\}$. Let $\mathcal{T}$ denote the standard topology $\mathbb{R}$.

a) For each $N \in \mathbb{N}$, determine the closure, the limit points, and the isolated points of A_N in $(\mathbb{R}, \mathcal{T}_f)$.

b) Determine the closure, the limit points, and the isolated points of A in $(\mathbb{R}, \mathcal{T}_f)$.

c) Let $N \in \mathbb{N}$ and let $a_1, \ldots, a_N \in \mathbb{R}$. Let $\mathcal{T}_{f,N}$ denote the subspace topology of $\mathcal{T}_f$ on A_N. Show: There exists a continuous map $f_N : (A_N, \mathcal{T}_{f,N}) \to (\mathcal{T}, \mathbb{R})$ such that $f(\frac{1}{n}) = a_n$ for $n = 1, \ldots, N$, $f(0) = 0$.

d) Show: There exists no continuous map $f : (\mathbb{R}, \mathcal{T}_f) \to (\mathcal{T}, \mathbb{R})$ such that $f(1) = 1$, $f(0) = 0$.

5. For $n \in \mathbb{N}$, let $f_n : [-\frac{1}{2}, \frac{1}{2}] \to \mathbb{R}$, $f_n(x) = n \tan(\frac{x}{n})$.

a) Show that the sequence of functions (f_n) converges pointwise on $[-\frac{1}{2}, \frac{1}{2}]$, and determine the limit function f.

b) Show that the family $\{f_n : n \in \mathbb{N}\} \subset \mathcal{C}([-\frac{1}{2}, \frac{1}{2}], \mathbb{R})$ is pointwise bounded and equicontinuous on $[-\frac{1}{2}, \frac{1}{2}]$.

c) Show that (f_n) converges uniformly to the limit function f on $[-\frac{1}{2}, \frac{1}{2}]$.